



Efe Demirer

Candidate RF & Microwave Systems Engineer · Antennas & Test Automation

Senior Electrical and Electronics Engineering student at Yaşar University specializing in RF & Microwave systems, antenna radiation pattern testing, and high-frequency instrumentation automation. Laboratory and defense experience at ASELSAN REHİS Test Centers Directorate conducting planar near-field antenna scanning, VNA 2-port calibrations, RF amplifier characterization, and real-time spectrum analysis.

[Live Portfolio \(GitHub Pages\)](#)[LinkedIn Profile](#)efedemirer.tr@gmail.com[GitHub: efedemirerr](#)

SKILLS & HIGHLIGHTS

Core RF & Microwave Highlights · ASELSAN REHİS

[ASELSAN REHİS RF Intern](#)[Antennas & Near-Field Scanning](#)[VNA & Spectrum Automation](#)[P1dB Amplifier Compression](#)[Antennas & Propagation](#)[VNA 2-Port Calibration](#)[Near-Field Anechoic Chambers](#)[RF Power Amplifiers \(P1dB\)](#)[NRL Arch Testing \(Absorbers\)](#)[C# .NET Test Automation](#)

RF & Microwave Engineering

[Antennas & Propagation](#)[Vector Network Analyzers \(VNA\)](#)[Real-Time Spectrum Analyzers](#)[VNA Full 2-Port Calibration](#)[P1dB Compression Testing](#)[RF Component Characterization](#)[Near-Field Anechoic Testing](#)[NRL Arch Absorber Testing](#)[S-Parameters & Impedance Matching](#)

RF Test Automation & Software

[C#](#)[.NET](#)[Windows Forms](#)[Python](#)[API Integration](#)[SCPI & G-Code](#)[Serial \(RS-232\)](#)[Ethernet TCP/IP](#)[MATLAB](#)

Motion & Scanning Hardware

[Cartesian X-Y Scanners](#)[Stepper Motor Control](#)[CNC Shield V3](#)[GRBL Firmware](#)[Limit Switch Safety Interlocks](#)[Arduino & STM32](#)

Electrical & Energy Systems

[Solar PV Design \(GES\)](#)[PVsyst Simulation](#)[ETAP Power Analysis](#)[AutoCAD](#)[Reactive Power Compensation](#)[High Voltage Inspection](#)[RCD & Grounding Tests](#)

Simulation & Tools

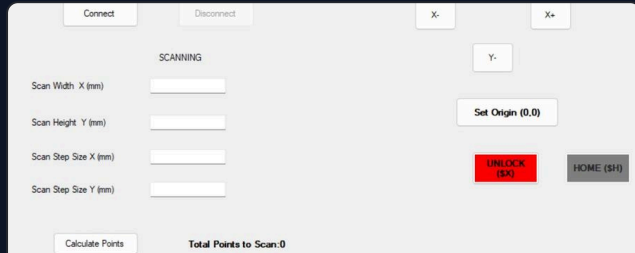
[Simulink](#)[LTspice](#)[Proteus](#)[Git / GitHub](#)[Visual Studio](#)[Microsoft Project](#)

FEATURED RF & MICROWAVE DEFENSE PROJECTS

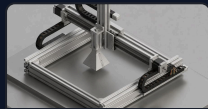
ASELSAN REHİS · ANTENNA TEST AUTOMATION

Automated CNC Antenna Scanning Test System & Control GUI

Anechoic Chamber Planar Near-Field Testing · 2026



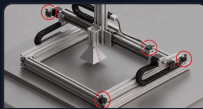
EFEScan C# .NET Control GUI · Live Scan Grid Coordinate Tracking



Cartesian Rail Chassis



CNC Shield V3 Hardware



4x NC Safety Interlocks

Motion: Cartesian X-Y (GRBL)**Safety:** 4x NC Limit Switches**Interface:** Serial COM (115200)**Application:** Near-Field Chamber

Autonomous Cartesian X-Y planar antenna scanning system engineered for ASELSAN REHİS Test Centers Directorate, replacing manual probe positioning with high-precision automated radiation pattern measurements in anechoic chambers.

- **Hardware & Motion Control:** Interfaced PC with Arduino Uno + CNC Shield V3 running GRBL firmware over asynchronous Serial Port (115200 Baud).
- **G-Code Protocol:** Implemented real-time G-Code stream generation (G90/G91/G92/G1) with centered grid coordinate pathway calculations and dynamic step resolution.
- **Hardware Safety Interlocks:** Integrated 4 mechanical Normally-Closed (NC) limit switches for hardware-level collision prevention, triggering instant GRBL ALARM mode and motor cutoff.
- **Multi-Threaded Execution:** Designed asynchronous BackgroundWorker threads eliminating UI freezing, paired with live bitmap scan visualization and feedrate-calibrated motion delays.

C# .NET

GRBL G-Code

Arduino & CNC Shield

Hardware Limit Switches

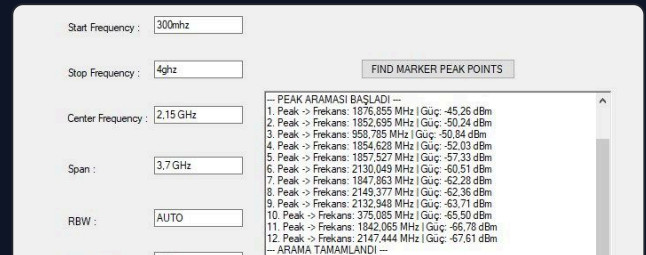
Near-Field Scanning

Anechoic Chamber

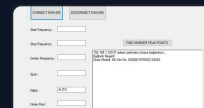
ASELSAN REHİS · RF SIGNAL ANALYSIS & INSTRUMENTATION

Real-Time Spectrum Analyzer API & Custom GUI

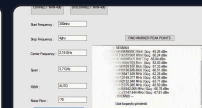
High-Speed Ethernet API & Real-Time Peak Detection · 2026



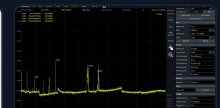
Custom C# GUI Real-Time Spectrum Trace & Active Peak Detection



Ethernet TCP API Flow



Smart RBW / VBW Cont...



Reference Validation

Protocol: Ethernet TCP/IP (Port 5000)**Algorithm:** Real-Time Peak Search**Noise Floor:** Dynamic -70 dBm FL...**Validation:** Reference Benchmarked

Dedicated C# desktop automation platform and custom GUI interfacing natively with a Real-Time Spectrum Analyzer at ASELSAN REHİS, executing automated high-speed RF spectral signal monitoring and carrier tracking.

- **Native API Integration:** Interfaced natively via manufacturer C/C++ API over Ethernet TCP/IPv4 (Port 5000) with robust IP parsing and persistent hardware session control.
- **Parameter Management:** Implemented dynamic frequency range configuration (Start/Stop, Center, Span) with automated Resolution Bandwidth (RBW) and Video Bandwidth (VBW) tuning.
- **Proprietary Peak Search:** Designed real-time peak detection isolating carrier frequencies above calibrated noise floor (-70 dBm) to separate true signals from ambient RF noise.
- **Laboratory Benchmarking:** Conducted comparative validation against manufacturer reference software, verifying measurement accuracy in defense testing environments.

C# .NET

Spectrum Analyzer API

Ethernet TCP/IP

Peak Search Algorithm

Spectral Monitoring

RF Sniffing

TELECOM & AUTONOMOUS TELEMETRY PROJECTS

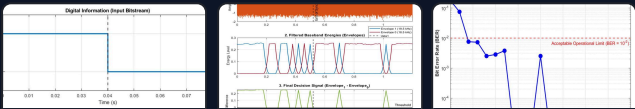
DIGITAL COMMUNICATIONS & DSP RESEARCH

Ultrasonic Ticketing System: Seamless & Offline Access Control

Near-Ultrasound Acoustic Communications · MATLAB DSP · 2026



Near-Ultrasound Acoustic Communication System Architecture



BFSK Modulated Carrier

Quadrature Mixer & DSP

BER / PSR Waterfall

Acoustic Band: 18 kHz – 22 kHz

Modulation: BFSK + Barker Sync

Channel: Two-Ray Multipath & D...

Platform: Full MATLAB Pipeline

Near-ultrasound (18–22 kHz) acoustic communication protocol enabling offline, zero-infrastructure access control without cellular data, internet, Bluetooth, or NFC, using standard commercial audio transducers.

- **Acoustic Transducers:** Utilized commercial smartphone speakers and microphones to transmit encrypted access tokens completely inaudible to human ears.
- **Digital Modulation:** Developed full MATLAB DSP pipeline featuring bit-level token synthesis, Barker code frame synchronization, and Binary Frequency Shift Keying (BFSK).
- **Channel Simulation:** Modeled multipath propagation (Two-Ray Model), ambient acoustic interference, transducer frequency limits, and motion-induced Doppler shifts.
- **DSP Receiver & Validation:** Built quadrature mixer receiver with envelope detection, parity verification, and comprehensive BER/PSR waterfall performance analysis.

MATLAB

Digital Communications

BFSK Modulation

Acoustic Channel Modeling

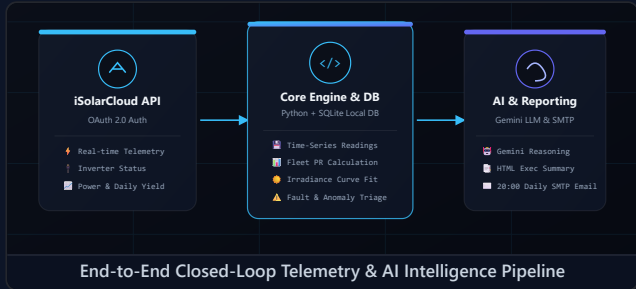
DSP

Barker Code

AUTONOMOUS TELEMETRY & AI SYSTEMS

SolarOPA: Autonomous AI Solar Plant Monitoring & Telemetry

Commercial PV Fleet Intelligence · Python & Gemini AI · 2026



End-to-End Closed-Loop Telemetry & AI Intelligence Pipeline



Telemetry: OpenAPI (OAuth 2.0)

Storage: SQLite Time-Series DB

Analytics: Scientific Fleet PR Curve

AI Engine: Gemini LLM (Daily SMT...

Autonomous closed-loop telemetry and operational intelligence platform for commercial solar power plants (GES) utilizing Sungrow iSolarCloud infrastructure, automating multi-inverter monitoring and anomaly diagnosis.

- **OpenAPI Telemetry:** Connected to Sungrow OpenAPI via OAuth 2.0 token management, streaming real-time multi-inverter generation, DC/AC voltages, and grid status.
- **Local Database Architecture:** Architected local SQLite time-series database storing minute-by-minute telemetry, inverter mappings, and grid fault histories.
- **Fleet PR Modeling:** Computed scientific Fleet Performance Ratio (PR) modeled against dynamic solar irradiance curves while filtering out plant anomalies.
- **Gemini AI Integration:** Prompt-engineered Google Gemini AI to analyze multi-point telemetry, diagnose anomalies, and compile executive HTML reports delivered daily at 20:00 via SMTP email.

Python

REST API

OAuth 2.0

SQLite DB

Google Gemini AI

Solar Telemetry

PROFESSIONAL ENGINEERING EXPERIENCE

Intern Engineer (RF & Microwave Systems / Test Automation) · ASELSAN REHİS Test Centers Directorate

06/2026 – 08/2026

Ankara, Turkey · Electronic Warfare & Radar Test Centers Directorate · Anechoic Chamber & Microwave Facilities

- **Planar Near-Field Antenna Scanning Automation:** Developed automated Cartesian X-Y motion control routines and C# .NET Windows Forms GUIs for anechoic chamber antenna measurements.
- **VNA 2-Port Calibration & Reference Extension:** Performed full 2-port SOLT calibrations on Vector Network Analyzers, extending measurement reference planes to test cable ends.
- **RF Power Amplifier Characterization:** Measured amplifier gain and executed P1dB 1-dB compression point tests to evaluate saturation behavior and maximum safe operating power.
- **Near-Field Chamber & NRL Arch Testing:** Conducted near-field anechoic chamber measurements and evaluated radar-absorbent material (RAM) reflectivity using NRL Arch testing.
- **Real-Time Spectrum Analyzer API & Signal Sniffing:** Integrated C/C++ API communication over Ethernet TCP/IPv4, implemented automated RBW tuning and a real-time peak detection algorithm (-70 dBm).
- **Hardware Interfacing & Safety Interlocks:** Configured Arduino Uno + CNC Shield V3 with GRBL firmware, driving stepper motors with 4x NC limit switch fail-safe cutoffs.

Intern Engineer · OPA Mühendislik

07/2025 – 08/2025 & 08/2026 – 09/2026

İzmir, Turkey · High Voltage Facilities & Solar PV Engineering

- **Solar PV Electrical Engineering:** Prepared AutoCAD single-line diagrams, electrical layout projects, and production yield simulations using PVsyst and ETAP.
- **Testing & Safety Inspections:** Conducted earth grounding resistance measurements and residual current device (RCD) tripping tests under High Voltage Operating Responsibility.
- **Reactive Power Factor Compensation:** Designed compensation panels, sized capacitor banks, and tuned reactive power relays to maintain zero reactive penalty.
- **EV Charging Infrastructure:** Calculated cable sizing, voltage drop limits, and electrical panel protections for high-power commercial EV chargers.

EDUCATION & ACADEMIC HONORS

Yaşar University – İzmir

09/2022 – Present

B.Sc. in Electrical and Electronics Engineering · 100% Merit Scholarship · GPA: 3.73 / 4.00 (High Honor Standing)

- **Core RF & Telecom Coursework:** Antennas and Propagation (EEE 4340), Electromagnetic Field Theory, Microwave & RF Measurement, Digital Communications, Signals and Systems.
- **Academic Honors:** Ranked top in department with full merit scholarship across all academic semesters.

Cem Bakioğlu Anatolian High School

09/2018 – 06/2022

High School Education · Salutatorian (Ranked 2nd in School) · Graduation Grade: 96.89 / 100

- **Academic Focus:** Advanced Mathematics, Physics & Analytical Problem Solving.
- **Science Honors:** TÜBİTAK 4006 Science and Society Fair Project Lead.

RF & DEFENSE ENGINEERING TECHNICAL COMPETENCIES

RF & Microwave Testing

- VNA Full 2-Port SOLT Calibration
- Reference Plane Shifting to Cables
- Amplifier P1dB Compression Tests
- NRL Arch Radar Absorber Testing
- Anechoic Chamber Measurements

Motion & Embedded Systems

- Planar Cartesian X-Y Motion Control
- Arduino Uno & CNC Shield V3
- GRBL Firmware & G-Code Automation
- 4x NC Limit Switch Safety Loops
- Stepper Motor Driver Tuning

Instrumentation & Software

- C# / .NET Windows Forms GUIs
- Ethernet TCP/IPv4 Socket Streaming
- Real-Time Peak Search (-70 dBm)
- Multi-Threaded BackgroundWorkers
- MATLAB Signal Processing & DSP

CERTIFICATIONS & CREDENTIALS

- National Technology Academy:** Chip Design (Çip Tasarımı) Specialization Certificate (Code: 35867118094667) · Digital ASIC & Semiconductor Design
- ASELSAN:** "a Yetenek" Engineering Internship Program Certificate of Completion (2026) · Defense RF Systems
- ROKETSAN:** LEVEL UP AI – Smart Object Detection and Tracking with YOLO (2026) · Edge AI
- ROKETSAN:** LEVEL UP AI – Autonomous AI Agent Development & Document Assistants (2026)
- TUSAŞ (TAI):** LIFT UP Industry-Oriented Capstone Projects Program (2025) · Defense Capstone
- Savunma Sanayii Akademi (SSB):** Milli Yetkinlik Hamlesi Competency & Career Summit (2026)
- MathWorks:** Core Signal Processing Techniques in MATLAB (Spectral Analysis, Filter Design)
- MathWorks:** Simulink Fundamentals (Model-Based Simulation) & Simscape Physical Systems
- MathWorks:** Signal Processing Onramp & MATLAB Onramp (2025)
- TÜBİTAK:** 4006 Science and Society Support Program Project Lead

LANGUAGES & INDUSTRY STATUS

- Turkish:** Native Proficiency
- English:** Professional Working Proficiency (B2 Upper-Intermediate, SFL Certified)
- Driving License:** Class B Active Driver
- Military Service:** Postponed until 31/12/2031
- Safety Credentials:** ASELSAN Occupational Health & Safety (İSG) Certified
- Internship Availability:** 2027 Spring Semester (01.02.2027 – 22.05.2027), minimum 3 full days/week on-site

Defense Industry Readiness: Fully cleared for high-frequency test laboratory and anechoic chamber environments with high-voltage and occupational safety compliance, adaptable to defense project sprints and on-site testing requirements.

PROFESSIONAL REFERENCES

Prof. Dr. Mustafa SEÇMEN

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Academic & Technical Verification: This portfolio represents verified engineering coursework, defense laboratory internships at ASELSAN REHİS Test Centers Directorate, and original technical implementations engineered by Efe Demirer. All measurement datasets, GUI software, and DSP simulation models are available for technical demonstration upon request.